

NaiveBayes

tions...

Stop

11:33:22 - bayes.NaiveBayes

```

a  b  c  <-- classified as
191 0  7 |  a = Apple
  0 180 0 |  b = Banana
 15 0 185 |  c = Orange

```

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseNaiveBayes

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

Classifier output

Relation: fruit_classification_rigorous

Instances: 578

Attributes: 5

weight

roundness

redness

yellowness

class

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

Attribute	Apple	Banana	Orange
	(0.34)	(0.31)	(0.35)

=====

weight

mean	151.0494	123.9346	160.6957
std. dev.	7.8577	10.3647	10.1811
weight sum	198	180	200
precision	1.0615	1.0615	1.0615

roundness

mean	8.1234	2.3857	8.7429
std. dev.	0.9271	0.7917	0.9621
weight sum	198	180	200
precision	1.2857	1.2857	1.2857

redness

mean	6.2525	1.65	5.875
std. dev.	2.7205	0.9097	1.1659
weight sum	198	180	200
precision	1	1	1

yellowness

mean	4.0253	7.9611	8.745
std. dev.	2.2303	1.439	0.8718
weight sum	198	180	200
precision	1	1	1

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	556	96.1938 %
Incorrectly Classified Instances	22	3.8062 %
Kappa statistic	0.9428	
Mean absolute error	0.0343	
Root mean squared error	0.1316	
Relative absolute error	7.7282 %	
Root relative squared error	27.9357 %	
Total Number of Instances	578	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.965	0.039	0.927	0.965	0.946	0.917	0.996	0.993	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.925	0.019	0.964	0.925	0.944	0.916	0.996	0.993	Orange
Weighted Avg.	0.962	0.020	0.962	0.962	0.962	0.942	0.997	0.995	

=== Confusion Matrix ===

a	b	c	-- classified as
191	0	7	a = Apple
0	180	0	b = Banana
15	0	185	c = Orange

Status

OK

Log

x0

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseNaiveBayes

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds2

%66

More options...

(Nom) class

StartStop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

Classifier output

Relation: fruit_classification_rigorous

Instances: 578

Attributes: 5

weight

roundness

redness

yellowness

class

Test mode: 2-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

Attribute	Apple	Banana	Orange
	(0.34)	(0.31)	(0.35)

=====

weight

mean	151.0494	123.9346	160.6957
std. dev.	7.8577	10.3647	10.1811
weight sum	198	180	200
precision	1.0615	1.0615	1.0615

roundness

mean	8.1234	2.3857	8.7429
std. dev.	0.9271	0.7917	0.9621
weight sum	198	180	200
precision	1.2857	1.2857	1.2857

redness

mean	6.2525	1.65	5.875
std. dev.	2.7205	0.9097	1.1659
weight sum	198	180	200
precision	1	1	1

yellowness

mean	4.0253	7.9611	8.745
std. dev.	2.2303	1.439	0.8718
weight sum	198	180	200
precision	1	1	1

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	556	96.1938 %
Incorrectly Classified Instances	22	3.8062 %
Kappa statistic	0.9428	
Mean absolute error	0.0348	
Root mean squared error	0.1327	
Relative absolute error	7.8378 %	
Root relative squared error	28.1603 %	
Total Number of Instances	578	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.970	0.042	0.923	0.970	0.946	0.917	0.996	0.992	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.920	0.016	0.968	0.920	0.944	0.916	0.996	0.952	Orange
Weighted Avg.	0.962	0.020	0.963	0.962	0.962	0.942	0.997	0.995	

=== Confusion Matrix ===

a	b	c	<-- classified as
192	0	6	a = Apple
0	180	0	b = Banana
16	0	184	c = Orange

StatusOK

Logx1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseNaiveBayes

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds8

%66

More options...

(Nom) class

StartStop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

Classifier output

Relation: fruit_classification_rigorous

Instances: 578

Attributes: 5

weight

roundness

redness

yellowness

class

Test mode: 8-fold cross-validation

=== Classifier model (full training set) ===

Naive Bayes Classifier

Attribute	Apple	Banana	Orange
	(0.34)	(0.31)	(0.35)

=====

weight

mean	151.0494	123.9346	160.6957
std. dev.	7.8577	10.3647	10.1811
weight sum	198	180	200
precision	1.0615	1.0615	1.0615

roundness

mean	8.1234	2.3857	8.7429
std. dev.	0.9271	0.7917	0.9621
weight sum	198	180	200
precision	1.2857	1.2857	1.2857

redness

mean	6.2525	1.65	5.875
std. dev.	2.7205	0.9097	1.1659
weight sum	198	180	200
precision	1	1	1

yellowness

mean	4.0253	7.9611	8.745
std. dev.	2.2303	1.439	0.8718
weight sum	198	180	200
precision	1	1	1

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	554	95.8478 %
Incorrectly Classified Instances	24	4.1522 %
Kappa statistic	0.9377	
Mean absolute error	0.0353	
Root mean squared error	0.1352	
Relative absolute error	7.944 %	
Root relative squared error	28.7044 %	
Total Number of Instances	578	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.960	0.042	0.922	0.960	0.941	0.909	0.996	0.992	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.920	0.021	0.958	0.920	0.939	0.908	0.996	0.992	Orange
Weighted Avg.	0.958	0.022	0.959	0.958	0.958	0.937	0.997	0.994	

=== Confusion Matrix ===

a	b	c	<-- classified as
190	0	8	a = Apple
0	180	0	b = Banana
16	0	184	c = Orange

Status

OK

Log

x1

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseMultilayerPerceptron-L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H a

Test options

Use training set

Supplied test set

Cross-validation

Percentage split

Folds10

%66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

Classifier output

Sigmoid Node 1

InputsWeights

Threshold-7.4150155548751905

Node 33.60424936378718

Node 40.5158622308396034

Node 58.020904860193431

Sigmoid Node 2

InputsWeights

Threshold-5.746016566824306

Node 3-9.358565426690792

Node 411.608367262589596

Node 5-4.372680490119331

Sigmoid Node 3

InputsWeights

Threshold-2.344762362165868

Attrib weight-8.57649480118818

Attrib roundness-1.0780463914083797

Attrib roundness-3.006969893040201

Attrib redness-10.65352054320388

Attrib yellowness1.34955942597144

Sigmoid Node 4

InputsWeights

Threshold1.4601934740195588

Attrib weight-8.57649480118818

Attrib roundness-0.4668649612128316

Attrib redness8.763796940239402

Attrib yellowness16.943688788355548

Sigmoid Node 5

InputsWeights

Threshold-1.5203018097237522

Attrib weight-1.974171474056887

Attrib roundness-8.722360284619685

Attrib redness-1.2144013810133896

Attrib yellowness0.21599916331559688

Class Apple

Input

Node 0

Class Banana

Input

Node 1

Class Orange

Input

Node 2

Time taken to build model: 0.11 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances57499.308 %

Incorrectly Classified Instances40.692 %

Kappa statistic0.9896

Mean absolute error0.0117

Root mean squared error0.0691

Relative absolute error2.6369 %

Root relative squared error14.6691 %

Total Number of Instances578

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.985	0.003	0.995	0.985	0.990	0.985	0.995	0.996	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.985	0.008	0.985	0.985	0.990	0.985	1.000	0.999	Orange
Weighted Avg.	0.993	0.004	0.993	0.993	0.993	0.989	0.998	0.998	

=== Confusion Matrix ===

a	b	c	-- classified as
195	0	3	a = Apple
0	180	0	b = Banana
1	0	199	c = Orange

StatusOK

Logx1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H 8

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds 8

% 66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

Classifier output

Sigmoid Node 1

Inputs Weights

Threshold -7.4150155548751905

Node 3 3.60424936378718

Node 4 0.5158622308396034

Node 5 8.020904860193431

Sigmoid Node 2

Inputs Weights

Threshold -5.746016566824306

Node 3 -9.358565426690792

Node 4 11.608367262589596

Node 5 -4.372680490119331

Sigmoid Node 3

Inputs Weights

Threshold -2.344762362165868

Attrib weight -8.576494800118818

Attrib roundness -1.0780463914083797

Attrib roundness -3.006969893040201

Attrib redness -10.65352054320388

Attrib yellowness 1.34955942597144

Sigmoid Node 4

Inputs Weights

Threshold 1.4601934740198588

Attrib weight -8.576494800118818

Attrib roundness -0.4668649612128316

Attrib redness 8.763796940239402

Attrib yellowness 16.943688788355548

Sigmoid Node 5

Inputs Weights

Threshold -1.5203018097237522

Attrib weight -1.974171474056887

Attrib roundness -8.722360284619685

Attrib redness -1.2144013810133896

Attrib yellowness 0.21599916331559688

Class Apple

Input

Node 0

Class Banana

Input

Node 1

Class Orange

Input

Node 2

Time taken to build model: 0.08 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 574 99.308 %

Incorrectly Classified Instances 4 0.692 %

Kappa statistic 0.9896

Mean absolute error 0.0122

Root mean squared error 0.0707

Relative absolute error 2.7506 %

Root relative squared error 15.0086 %

Total Number of Instances 578

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.990	0.005	0.990	0.990	0.990	0.985	0.996	0.996	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.990	0.005	0.990	0.990	0.990	0.985	0.996	0.995	Orange
Weighted Avg.	0.993	0.004	0.993	0.993	0.993	0.989	0.998	0.997	

=== Confusion Matrix ===

a	b	c	-- classified as
196	0	2	a = Apple
0	180	0	b = Banana
2	0	198	c = Orange

Status

OK

Log

 x1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

MultilayerPerceptron -L 0.3 -M 0.2 -N 500 -V 0 -S 0 -E 20 -H *

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds 2

% 66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

Classifier output

Sigmoid Node 1

Inputs Weights

Threshold -7.4150155548751905

Node 3 3.60424936378718

Node 4 0.5158622308396034

Node 5 8.020904860193431

Sigmoid Node 2

Inputs Weights

Threshold -5.746016566824306

Node 3 -9.358565426690792

Node 4 11.608367262589596

Node 5 -4.372680490119331

Sigmoid Node 3

Inputs Weights

Threshold -2.344762362165868

Attrib weight -8.576494800118818

Attrib roundness -1.0780463914083797

Attrib roundness -3.006969893040201

Attrib redness -10.653592054320388

Attrib yellowness 1.34959942597144

Sigmoid Node 4

Inputs Weights

Threshold 1.4601934740195588

Attrib weight -8.576494800118818

Attrib roundness -0.4668649612128316

Attrib redness 8.763796940239402

Attrib yellowness 16.943688788355548

Sigmoid Node 5

Inputs Weights

Threshold -1.5203018097237522

Attrib weight -1.974171474056887

Attrib roundness -8.722360284619685

Attrib redness -1.2144013810133896

Attrib yellowness 0.21599916331559688

Class Apple

Input

Node 0

Class Banana

Input

Node 1

Class Orange

Input

Node 2

Time taken to build model: 0.08 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 572 98.9619 %

Incorrectly Classified Instances 6 1.0381 %

Kappa statistic 0.9844

Mean absolute error 0.0176

Root mean squared error 0.0802

Relative absolute error 3.9588 %

Root relative squared error 17.0242 %

Total Number of Instances 578

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.975	0.003	0.995	0.975	0.985	0.977	0.997	0.996	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.985	0.013	0.975	0.985	0.985	0.977	0.989	0.998	Orange
Weighted Avg.	0.990	0.005	0.990	0.990	0.990	0.984	0.995	0.998	

=== Confusion Matrix ===

a	b	c	-- classified as
193	0	5	a = Apple
0	180	0	b = Banana
1	0	199	c = Orange

Status

OK

Log

 x1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseSimpleLogistic -I 0 -M 500 -H 50 -W 0.0

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Folds2

%66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:35 - functions.SimpleLogistic

Classifier output

=== Run information ===

Scheme: weka.classifiers.functions.SimpleLogistic -I 0 -M 500 -H 50 -W 0.0

Relation: fruit_classification_rigorous

Instances: 578

Attributes: 5

weight

roundness

redness

yellowness

class

Test mode: 2-fold cross-validation

=== Classifier model (full training set) ===

SimpleLogistic:

Class Apple :

0.41 +

[weight] * 0.12 +

[redness] * -0.75 +

[yellowness] * -2.04

Class Banana :

32.38 +

[roundness] * -5.83

Class Orange :

-10.16 +

[weight] * -0.04 +

[roundness] * 0.32 +

[redness] * 1.19 +

[yellowness] * 1.28

Time taken to build model: 0.09 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances56798.0969 %

Incorrectly Classified Instances111.9031 %

Kappa statistic0.9714

Mean absolute error0.0288

Root mean squared error0.096

Relative absolute error6.4792 %

Root relative squared error20.378 %

Total Number of Instances578

=== Detailed Accuracy By Class ===

TP RateFP RatePrecisionRecallF-MeasureMCCROC AreaPRC AreaClass

0.9750.0160.9700.9750.9720.9580.9990.998Apple

1.0000.0001.0001.0001.0001.0001.0001.000Banana

0.9700.0130.9750.9700.9720.9580.9990.998Orange

Weighted Avg.0.9810.0100.9810.9810.9810.9710.9990.999

=== Confusion Matrix ===

a b c <-- classified as

193 0 5 | a = Apple

0 180 0 | b = Banana

6 0 194 | c = Orange

Status

OK

Log

x1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseSimpleLogistic -I 0 -M 500 -H 50 -W 0.0

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Folds8

%66

Set...More options...

(Nom) class

StartStop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:55 - functions.SimpleLogistic

12:00:43 - functions.SimpleLogistic

Classifier output

=== Run information ===

Scheme: weka.classifiers.functions.SimpleLogistic -I 0 -M 500 -H 50 -W 0.0

Relation: fruit_classification_rigorous

Instances: 578

Attributes: 5

weight

roundness

redness

yellowness

class

Test mode: 8-fold cross-validation

=== Classifier model (full training set) ===

SimpleLogistic:

Class Apple :

0.41 +

[weight] * 0.12 +

[redness] * -0.75 +

[yellowness] * -2.04

Class Banana :

32.38 +

[roundness] * -5.83

Class Orange :

-10.16 +

[weight] * -0.04 +

[roundness] * 0.32 +

[redness] * 1.19 +

[yellowness] * 1.28

Time taken to build model: 0.04 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances56998.4429 %

Incorrectly Classified Instances91.5571 %

Kappa statistic0.9766

Mean absolute error0.0205

Root mean squared error0.0878

Relative absolute error4.6092 %

Root relative squared error18.6435 %

Total Number of Instances578

=== Detailed Accuracy By Class ===

TP RateFP RatePrecisionRecallF-MeasureMCCROC AreaPRC AreaClass

0.9800.0130.9750.9800.9770.9650.9990.998Apple

1.0000.0001.0001.0001.0001.0001.0001.000Banana

0.9750.0110.9800.9750.9770.9660.9990.998Orange

Weighted Avg.0.9840.0080.9840.9840.9840.9760.9990.999

=== Confusion Matrix ===

a b c <-- classified as

194 0 4 | a = Apple

0 180 0 | b = Banana

5 0 195 | c = Orange

Status

OK

Log

x1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseSimpleLogistic -I 0 -M 500 -H 50 -W 0.0

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Folds10

%66

Set...

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:55 - functions.SimpleLogistic

12:00:43 - functions.SimpleLogistic

12:01:04 - functions.SimpleLogistic

Classifier output

=== Run information ===

Scheme: weka.classifiers.functions.SimpleLogistic -I 0 -M 500 -H 50 -W 0.0

Relation: fruit_classification_rigorous

Instances: 578

Attributes: 5

weight

roundness

redness

yellowness

class

Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

SimpleLogistic:

Class Apple :

0.41 +

[weight] * 0.12 +

[redness] * -0.75 +

[yellowness] * -2.04

Class Banana :

32.38 +

[roundness] * -5.83

Class Orange :

-10.16 +

[weight] * -0.04 +

[roundness] * 0.32 +

[redness] * 1.19 +

[yellowness] * 1.28

Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances57298.9619 %

Incorrectly Classified Instances61.0381 %

Kappa statistic0.9844

Mean absolute error0.0154

Root mean squared error0.0795

Relative absolute error3.4649 %

Root relative squared error16.872 %

Total Number of Instances578

=== Detailed Accuracy By Class ===

TP RateFP RatePrecisionRecallF-MeasureMCCRoc AreaPRC AreaClass

0.9850.0080.9850.9850.9850.9770.9990.999Apple

1.0000.0001.0001.0001.0001.0001.0001.000Banana

0.9850.0080.9850.9850.9850.9770.9990.999Orange

Weighted Avg.0.9900.0050.9900.9900.9900.9841.0000.999

=== Confusion Matrix ===

a b c <-- classified as

195 0 3 | a = Apple

0 180 0 | b = Banana

3 0 197 | c = Orange

Status

OK

Log

x1

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseClassificationViaRegression -W weka.classifiers.trees.MSP -- -M 4.0 -num-decimal-places 4

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:55 - functions.SimpleLogistic

12:00:43 - functions.SimpleLogistic

12:01:04 - functions.SimpleLogistic

12:02:26 - meta.ClassificationViaRegression

Classifier output

LM num: 1

class =

-0.0113 * roundness

+ 1.0212

LM num: 2

class =

-0.0053 * roundness

+ 0.0464

Number of Rules : 2

Classifier for class with index 2:

M5 pruned model tree:

(using smoothed linear models)

redness <= 3.5 : LM1 (230/0%)

redness > 3.5 :

| yellowness <= 6.5 : LM2 (148/0%)

| yellowness > 6.5 : LM3 (200/0%)

LM num: 1

class =

0.0085 * redness

+ 0.0103 * yellowness

- 0.0896

LM num: 2

class =

0.0058 * redness

+ 0.0205 * yellowness

- 0.0942

LM num: 3

class =

0.0058 * redness

+ 0.0172 * yellowness

+ 0.8058

Number of Rules : 3

Time taken to build model: 0.11 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances

578

100

%

Incorrectly Classified Instances

0

0

%

Kappa statistic

1

Mean absolute error

0.0122

Root mean squared error

0.0196

Relative absolute error

2.7526 %

Root relative squared error

4.1587 %

Total Number of Instances

578

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Orange
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a

b

c

<-- classified as

198

0

0

|

a = Apple

0

180

0

|

b = Banana

0

0

200

|

c = Orange

Status

OK

Log

x1

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseClassificationViaRegression -W weka.classifiers.trees.MSP --M 4.0 -num-decimal-places 4

Test options

☐ Use training set

☐ Supplied test set

Set...

☒ Cross-validation

Folds 8

☐ Percentage split

% 66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:55 - functions.SimpleLogistic

12:00:43 - functions.SimpleLogistic

12:01:04 - functions.SimpleLogistic

12:02:26 - meta.ClassificationViaRegression

12:02:49 - meta.ClassificationViaRegression

Classifier output

LM num: 1

class =

-0.0113 * roundness

+ 1.0212

LM num: 2

class =

-0.0053 * roundness

+ 0.0464

Number of Rules : 2

Classifier for class with index 2:

M5 pruned model tree:

(using smoothed linear models)

redness <= 3.5 : LM1 (230/0%)

redness > 3.5 :

| yellowness <= 6.5 : LM2 (148/0%)

| yellowness > 6.5 : LM3 (200/0%)

LM num: 1

class =

0.0085 * redness

+ 0.0103 * yellowness

- 0.0896

LM num: 2

class =

0.0058 * redness

+ 0.0205 * yellowness

- 0.0942

LM num: 3

class =

0.0058 * redness

+ 0.0172 * yellowness

+ 0.8058

Number of Rules : 3

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances

578

100

%

Incorrectly Classified Instances

0

0

%

Kappa statistic

1

Mean absolute error

0.0122

Root mean squared error

0.0188

Relative absolute error

2.7542 %

Root relative squared error

3.9985 %

Total Number of Instances

578

=== Detailed Accuracy By Class ===

TP Rate

FP Rate

Precision

Recall

F-Measure

MCC

ROC Area

PRC Area

Class

1.000

0.000

1.000

1.000

1.000

1.000

1.000

1.000

Apple

1.000

0.000

1.000

1.000

1.000

1.000

1.000

1.000

Banana

1.000

0.000

1.000

1.000

1.000

1.000

1.000

1.000

Orange

Weighted Avg.

1.000

0.000

1.000

1.000

1.000

1.000

1.000

1.000

=== Confusion Matrix ===

a

b

c

<-- classified as

198

0

0

a = Apple

0

180

0

b = Banana

0

0

200

c = Orange

Status

OK

Log

x1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

MultiClassClassifier -M 0 -R 2.0 -S 1 -W weka.classifiers.functions.Logistic --R 1.0E-8 -M -1 -num-decimal-places 4

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds 10

% 66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:55 - functions.SimpleLogistic

12:00:43 - functions.SimpleLogistic

12:01:04 - functions.SimpleLogistic

12:02:26 - meta.ClassificationViaRegression

12:02:49 - meta.ClassificationViaRegression

12:03:08 - meta.ClassificationViaRegression

12:03:35 - meta.MultiClassClassifier

Classifier output

roundness 13.4755

redness 0.0013

yellowness -1.725

Intercept -210.8781

Odds Ratios...

Class

Variable neg_Banana

weight 3.0366

roundness 711737.5742

redness 1.0013

yellowness 0.1782

Classifier 3, using indicator values: Strings: 3

Invert: false

Cols: 3

Logistic Regression with ridge parameter of 1.0E-8

Coefficients...

Class

Variable neg_Orange

weight 0.294

roundness -2.8982

redness -3.6193

yellowness -5.1194

Intercept 32.2624

Odds Ratios...

Class

Variable neg_Orange

weight 1.3418

roundness 0.0551

redness 0.0268

yellowness 0.006

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 570 99.6159 %

Incorrectly Classified Instances 8 1.3841 %

Kappa statistic 0.9792

Mean absolute error 0.0164

Root mean squared error 0.0842

Relative absolute error 3.7014 %

Root relative squared error 17.8647 %

Total Number of Instances 578

=== Detailed Accuracy By Class ===

TP Rate FP Rate Precision Recall F-Measure MCC ROC Area PRC Area Class

0.985 0.013 0.975 0.985 0.980 0.969 0.999 0.998 Apple

1.000 0.000 1.000 1.000 1.000 1.000 1.000 1.000 Banana

0.975 0.008 0.985 0.975 0.980 0.969 0.999 0.998 Orange

Weighted Avg. 0.986 0.007 0.986 0.986 0.986 0.979 0.999 0.998

=== Confusion Matrix ===

a b c <-- classified as

195 0 3 | a = Apple

0 180 0 | b = Banana

5 0 195 | c = Orange

Status

OK

Log

x1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

Choose

MultiClassClassifier -M 0 -R 2.0 -S 1 -W weka.classifiers.functions.Logistic --R 1.0E-8 -M -1 -num-decimal-places 4

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds 8

% 66

More options...

(Nom) class

Start

Stop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:55 - functions.SimpleLogistic

12:00:43 - functions.SimpleLogistic

12:01:04 - functions.SimpleLogistic

12:02:26 - meta.ClassificationViaRegression

12:02:49 - meta.ClassificationViaRegression

12:03:08 - meta.ClassificationViaRegression

12:03:35 - meta.MultiClassClassifier

12:03:55 - meta.MultiClassClassifier

Classifier output

roundness 13.4755

redness 0.0013

yellowness -1.725

Intercept -210.8781

Odds Ratios...

Class

Variable neg_Banana

=====

weight 3.0366

roundness 711737.5742

redness 1.0013

yellowness 0.1782

Classifier 3, using indicator values: Strings: 3

Invert: false

Cols: 3

Logistic Regression with ridge parameter of 1.0E-8

Coefficients...

Class

Variable neg_Orange

=====

weight 0.294

roundness -2.8982

redness -3.6193

yellowness -5.1194

Intercept 32.2624

Odds Ratios...

Class

Variable neg_Orange

=====

weight 1.3418

roundness 0.0551

redness 0.0268

yellowness 0.006

Time taken to build model: 0 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances 570 99.6159 %

Incorrectly Classified Instances 8 1.3841 %

Kappa statistic 0.9792

Mean absolute error 0.016

Root mean squared error 0.0853

Relative absolute error 3.6064 %

Root relative squared error 16.0938 %

Total Number of Instances 578

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.985	0.013	0.975	0.985	0.980	0.969	0.999	0.998	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.975	0.008	0.985	0.975	0.980	0.969	0.998	0.998	Orange
Weighted Avg.	0.986	0.007	0.986	0.986	0.986	0.979	0.999	0.998	

=== Confusion Matrix ===

a b c

<-- classified as

195 0 3 | a = Apple

0 180 0 | b = Banana

5 0 195 | c = Orange

Status

OK

Log

x1

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseMultiClassClassifier -M 0 -R 2.0 -S 1 -W weka.classifiers.functions.Logistic -- -R 1.0E-8 -M -1 -num-decimal-places 4

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Folds2

%66

More options...

(Nom) class

StartStop

Result list (right-click for options)

11:33:22 - bayes.NaiveBayes

11:55:23 - bayes.NaiveBayes

11:55:43 - bayes.NaiveBayes

11:57:25 - functions.MultilayerPerceptron

11:58:03 - functions.MultilayerPerceptron

11:59:14 - functions.MultilayerPerceptron

11:59:55 - functions.SimpleLogistic

12:00:43 - functions.SimpleLogistic

12:01:04 - functions.SimpleLogistic

12:02:26 - meta.ClassificationViaRegression

12:02:49 - meta.ClassificationViaRegression

12:03:08 - meta.ClassificationViaRegression

12:03:35 - meta.MultiClassClassifier

12:03:55 - meta.MultiClassClassifier

12:04:15 - meta.MultiClassClassifier

Classifier output

roundness13.4755
redness0.0013
yellowness-1.725
Intercept-210.8781

Odds Ratios...

Class

Variableneg_Banana

weight3.0366
roundness711737.5742
redness1.0013
yellowness0.1782

Classifier 3, using indicator values: Strings: 3

Invert: false

Cols: 3

Logistic Regression with ridge parameter of 1.0E-8

Coefficients...

Class

Variableneg_Orange

weight0.294
roundness-2.8982
redness-3.6193
yellowness-5.1194
Intercept32.2624

Odds Ratios...

Class

Variableneg_Orange

weight1.3418
roundness0.0551
redness0.0268
yellowness0.006

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances	569	98.4429 %
Incorrectly Classified Instances	9	1.5571 %
Kappa statistic	0.9766	
Mean absolute error	0.0157	
Root mean squared error	0.0829	
Relative absolute error	3.5341 %	
Root relative squared error	17.5857 %	
Total Number of Instances	578	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.985	0.016	0.970	0.985	0.977	0.966	0.998	0.997	Apple
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Banana
	0.970	0.008	0.985	0.970	0.977	0.966	0.998	0.997	Orange
Weighted Avg.	0.984	0.008	0.985	0.984	0.984	0.976	0.999	0.998	

=== Confusion Matrix ===

a	b	c	<-- classified as
195	0	3	a = Apple
0	180	0	b = Banana
6	0	194	c = Orange

Status
OK

Log x1